Updated Results Section

Full-grid reconciliation of the June 18 preliminary PDF. The main story survives: FC->SC is stronger than SC->FC, bv+demo is the required baseline, and observed FC is the only connectome representation that reliably adds cognition signal above that floor.

Two reporting fixes matter: use BayesianRidge/KernelRidge same-modality rows for Ceiling B, and state the estimator next to every absolute number.

- F1 replicates across Glasser and 4S456: FC->SC / SC->FC is 1.61x and 1.68x.
- F2 becomes a clean double dissociation: anatomy predicts SC, demographics predicts FC.
- F5 is clearest as an add-on test: obs_FC+bv+demo beats bv+demo for CogTotal and CogCryst; observed SC remains below the cheap baseline.
- F8 remains hypothesis-generating until the PC3 mechanism is checked on 4S456.

F1 and F2 - Reconstruction Spine

PCA->PLS reconstruction cells. Values are mean demeaned Pearson over 10 frozen seeds.

Parcellation	FC->SC	SC->FC	ratio
Glasser	0.136	0.084	1.61x
4S456Parcels	0.146	0.087	1.68x

F2 - Modality Dissociation

Anatomy wins for structural targets; demographics wins for functional targets.

Parcellation	bv->SC	demo->SC	bv->FC	demo->FC	bv+demo->SC	bv+demo->FC
Glasser	0.162	0.130	0.049	0.111	0.169	0.100
4S456Parcels	0.185	0.133	0.045	0.105	0.191	0.094

Ceiling B - Oracle Reconciliation

The preliminary SC->SC = 0.647 number matches BR/KR same-modality oracles, not bottlenecked PLS.

Parcellation	PLS SC->SC	BR SC->SC	KR SC->SC	BR FC->FC
Glasser	0.376	0.648	0.645	0.673
4S456Parcels	0.386	0.622	0.618	0.632

F3 - Downstream Imputation Asymmetry

BayesianRidge cognition rows. pred_SC improves over observed SC; pred_FC loses much of observed FC's cognition signal.

Parcellation	Target	pred_SC / obs_SC	pred_FC / obs_FC	utility asym
Glasser	CogTotal	1.37x	0.55x	2.49x
Glasser	CogFluid	1.41x	0.57x	2.48x
Glasser	CogCryst	1.36x	0.45x	3.03x
4S456Parcels	CogTotal	1.38x	0.61x	2.27x
4S456Parcels	CogFluid	1.66x	0.66x	2.52x
4S456Parcels	CogCryst	1.32x	0.52x	2.56x

F4 - Fraction Surviving bv+demo Removal

Residualized cognition signal divided by raw cognition score. FC survives; SC mostly collapses.

Parcellation	Input	Total	Fluid	Cryst
Glasser	obs_FC	67%	66%	78%
Glasser	obs_SC	16%	5%	24%
Glasser	pred_SC	49%	51%	57%
Glasser	pred_FC	17%	27%	-0%
4S456Parcels	obs_FC	65%	64%	74%
4S456Parcels	obs_SC	22%	19%	26%
4S456Parcels	pred_SC	51%	56%	54%
4S456Parcels	pred_FC	29%	43%	10%

F5 - Cost-Benefit Baseline

BayesianRidge downstream cognition. Lift is Pearson r minus the bv+demo baseline; p is the median paired-permutation p across seeds.

Parcellation	Input	Total r	Total lift	Total p	Cryst r	Cryst lift	Cryst p
Glasser	bv+demo	0.359	+0.000	1	0.354	+0.000	1
Glasser	obs_FC	0.447	+0.088	0.0802	0.487	+0.133	0.0307
Glasser	obs_SC	0.248	-0.111	0.939	0.253	-0.101	0.817
Glasser	obs_FC+bv+demo	0.474	+0.115	0.0157	0.516	+0.162	0.00225
Glasser	obs_SC+bv+demo	0.309	-0.050	0.824	0.319	-0.035	0.604
4S456Parcels	bv+demo	0.359	+0.000	1	0.354	+0.000	1
4S456Parcels	obs_FC	0.445	+0.086	0.0905	0.476	+0.122	0.0407
4S456Parcels	obs_SC	0.256	-0.103	0.927	0.259	-0.095	0.883
4S456Parcels	obs_FC+bv+demo	0.468	+0.109	0.0192	0.500	+0.146	0.008
4S456Parcels	obs_SC+bv+demo	0.308	-0.052	0.831	0.312	-0.042	0.695

New Takeaways and Write-up Posture

The completed grid is most useful because it turns the preliminary Glasser story into a cross-parcellation confirmatory spine, while cleanly separating confirmed findings from mechanism hypotheses.

- The asymmetry is not a target-reliability artifact: same-modality FC and SC oracles are closely matched under BR/KR.

- 4S456 specifically boosts ->SC prediction, suggesting finer structural detail rather than generic metric inflation.

- SC can be counterproductive downstream; obs_SC+bv+demo stays below bv+demo.

- pred_FC is actively harmful for cognition relative to bv+demo, sharpening non-substitutability.

- Use modest-effect language throughout: this is a methods and redirect paper, not a diagnostic biomarker paper.